

Design and analysis of omnidirectional mobile material handling robots that can adapt to complex environments

He Lei¹, Xu Canzhen², He Fuchun¹, Fu Chunming^{1,3,*}, Huo Shaoyong¹

¹School of Mechanical Engineering, University of South China, Hunan, Hengyang, 421200, China

²School of Electrical Engineering, University of South China, Hunan, Hengyang, 421200, China

³CNNC Nuclear Fuel Components Co., Ltd., Sichuan, Yibin, 644100, China

Abstract—A high-precision omnidirectional mobile material handling robot that can adapt to complex environments is designed, which can be applied to material handling in miscellaneous scenarios such as narrow compartments, dense cargo racks or obstacles. The handling robot is equipped with a set of (four) Mecanum wheels, which are controlled by motion through the STM32 system, and can be translated and rotated in any direction and autonomously planned in the path with the vision system. The robotic arm is mainly composed of three vertically staggered screw stepper motors and the bottom gimbal, with multiple chrome-plated light rods and linear bearings to achieve all-round, high-precision and long-distance operation. The design establishes a three-dimensional model of the handling robot, uses additive manufacturing (3D printing) technology to test and make the key part of the robotic arm - sliding seat, and performs strength check analysis on the part. Finally, through the physical grasping, handling and placement experiments in various complex environments, the high accuracy of the motion trajectory of the above robot and the reliability of the actuator are verified.

Keywords—robotic arm, material handling, Robot, McCannem wheel, Carry, automation

I. PREFACE

With the development of automation and intelligent technology, robots that can move autonomously have been used in more and more scenarios, and have become one of the current research hotspots. At present, it is mainly multi-legged, crawler, wheeled, wheeled, and orbital robots [1]. Among them, multi-legged and crawler robots can adapt to complex environmental landscapes, but their speed is slow and they are not suitable for high-efficiency handling scenarios; The wheel-foot robot has a large turning radius, which is not suitable for relatively narrow areas, and the movement is limited by the field; The moving route of the orbital robot is relatively fixed, and the transportation route is constrained by the track, which is not suitable for flexible working scenarios, while the omnidirectional mobile robot in the wheeled robot has the advantages of high efficiency and flexibility, and can operate in a small area without the restrictions between the fields, especially the omnidirectional mobile robot based on the McCham wheel can adapt to a variety of complex working environments. At present, the more advanced Macnum wheel-based robots are mainly used in medical, firefighting and autonomous cruise vending scenarios, while the design and research of handling robots is still in its infancy. Due to the rationality of the structure of the Mecanum wheel itself, its bearing capacity and sports

performance have good performance, so it is also more competent for various handling scenarios.

Among the various handling robots, the SCARA wafer handling robot with direct drive can achieve lightweight, high precision and smooth and reliable workpiece transfer, which is the mainstream structure of wafer handling in the future [2]. The multi-articulated transfer manipulator is mainly used for standardized container transfer in narrow aisles, where the space constraints are large and the transfer process requires high flexibility [3]. In the mode of small batch production of large aircraft parts, it is extremely unrealistic to design and develop different heavy-duty handling robots for hundreds of tooling, and it requires multi-robot systems to cooperate in handling [4]. The soft robotic arm has the characteristics of high redundancy and low stiffness, and the task space of the soft robotic arm is more complex and narrow than that of the rigid robotic arm, and it is difficult to realize the installation and calibration of machine vision equipment other than the robotic arm matrix [5-6]. Compared with ordinary six-axis robotic arm, the new conduit handling robot based on three-coordinate mobile platform and material transport rotating platform has larger working space and better working rigidity performance, and is suitable for the processing and manufacturing of aviation conduit by CNC pipe bending machine [7]. QJRB180-1 handling robot and the developed end effector can realize the grabbing, handling and stacking of luggage more efficiently and safely [8]. The robotic arm used in the handling robot has its own characteristics in terms of structure and control, and most of the robot's working capacity is limited to varying degrees because of the environment [9]. Germany's remote operation robot A1000 for handling nuclear waste is mainly composed of a driving system, a telescopic mechanism system, an operating arm system, and a control system [10]. It has good adaptability to harsh working conditions, and is easy to disassemble and maintain, but it is not flexible due to hydraulic drive, and can only be applied to fixed, large-scale handling environments, and the versatility is not high. The UR robot is a modular robot for teach-in developed in Denmark, which is small and flexible, and can be used on many complex teach-in work platforms [11]. However, its modular design has led to a decrease in its carrying capacity, so it is also more suitable for teaching than actual handling. The IRB460 articulated heavy-duty handling robot produced by ABB works stably and efficiently, but the energy dissipation of the joints of the robotic arm is too large under continuous heavy-duty work [12], and there are many factors limited by the working environment, which is not suitable for more flexible

intelligent handling scenarios.

In view of the above status quo and problems, an omnidirectional handling robot scheme based on the Mecanum wheel is proposed, and the movement of the robotic arm of the handling robot is decomposed at the same time, and the stepper motor and servo are used to ensure the high efficiency, high precision and high stability of the robot in the complex handling environment. The motion part of the robot using Cartesian coordinates consists of three moving parts perpendicular to each other. Its moving distance in each axis can directly read each coordinate axis, with high positioning accuracy and simple structure [2]. Considering that the larger volume of the robot is difficult to adapt to a variety of complex working environments and the flexibility is limited, this paper only takes the handling robot and $70 \times 50 \times 50 \text{ mm}^3$ materials in the volume of $300 \times 300 \times 400 \text{ mm}^3$ as an example, uses the 3D design software to establish the 3D model of the handling robot, and uses the finite element analysis software to analyze the key components of the robot. The first section mainly introduces the structural design of the omnidirectional mobile material handling robot, the second section analyzes the movement of the handling robot, the third section checks the strength of the key components of the robot and the physical production and verifies its function, and finally shows that the model is basically consistent with the test.

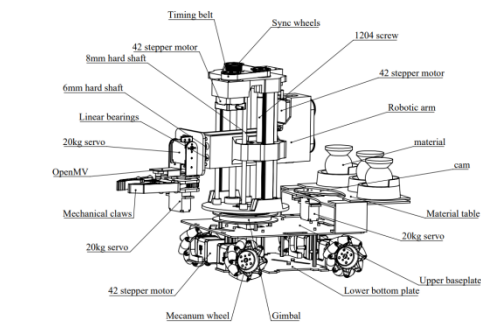
II. STRUCTURAL DESIGN OF OMNIDIRECTIONAL MOBILE MATERIAL HANDLING ROBOT

A. Application scenarios and design requirements

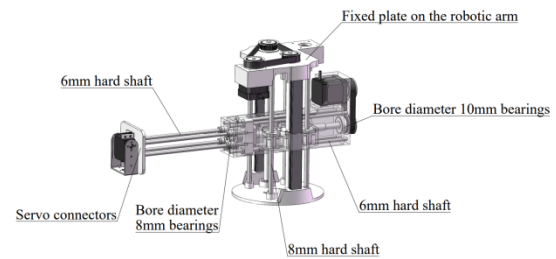
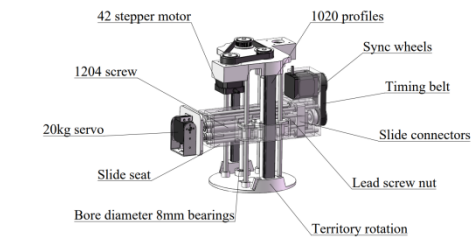
Compared with a variety of agricultural robots used in standardized agriculture and service robots in modern service industries, handling robots used in manufacturing have more extensive application scenarios, more complex environments, and stricter structural requirements and mechanical properties of robots. At the same time, the larger volume of the handling robot is only suitable for special environments, and the manufacturing and maintenance costs are high, so the $300 \times 300 \times 400 \text{ mm}^3$ volume requirements are selected as the design basis of the robot, and a handling robot with small volume, high flexibility and excellent mechanical properties is designed. This robot can be used not only for drug distribution, the handling of near-ground objects in factories, but also for the handling of higher objects with a variety of robots and conveyor belts, and even for the handling of small objects in the home environment.

B. Structure

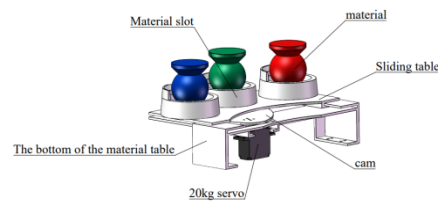
The overall structure diagram of the omnidirectional mobile handling robot is shown in Figure 1 (a), which is mainly composed of a Mecanum wheel, a robotic arm (screw, stepper motor, chrome-plated hard shaft, linear bearing, synchronous wheel timing belt and other components assembled by self-made parts), robotic claw, material table (retractable material device composed of servo and cam) and upper and lower bottom plates. The gimbal of the lower part of the robotic arm and the bottom plate of the vehicle are connected by copper pillars, the front end of the robotic arm is equipped with a self-made groove interface that can be connected with the robotic claw, the material table is connected by bolt nut connectors, the stepper motor and the Mecanum wheel are connected and fixed on the lower bottom plate through self-made connectors, and the upper and lower bottom plates are connected by copper pillars.



(a) The overall structure of the material handling robot



(b) Structure and telescopic diagram of the robotic arm



(c) The structure of the material table

Fig. 1 Overall structure and components of material handling robot

The detailed structure of the robotic arm and the material table is shown in Figure 1(b) and Figure 1(c), respectively. The overall structure of the robotic arm is inspired by the gantry structure, on the basis of the profiles on both sides as support, each side uses two 8mm chrome-plated hard shafts as a guide to minimize the jitter deviation when the robot arm moves, and is equipped with 8mm inner diameter flange linear bearings to increase lubrication and improve accuracy. Two 1204 screws are used in the vertical direction, and the inner diameter 8mm and 10mm rolling bearings are installed at both ends of the screw, which can avoid the unstable force of the robot arm when running up and down and improve stability. Connecting the support profiles are two homemade 3D prints, and the upper part of the prints connected to the profiles is designed with a mounting groove for a 42-stepper motor, which is fixed with four M3 bolts. The "hammock" that can be moved up and down, that is, the horizontal body of the robotic arm, is also made of homemade 3D prints, interacting vertically with vertical directions, and inside is a retractable structure made of 8 chrome-plated hard shafts plus a 3D print, supplemented by 8 linear bearings with an inner

diameter of 6mm to increase its accuracy and stability. When the stepper motor rotates, the synchronous wheel and the synchronous belt drive the screw located in the center of the 8 hard shafts to rotate (because the screw is limited to the print 1 (sliding seat) can only do rotational movement, and the nut above the screw can only move horizontally along the screw under the guidance of the 4 hard shafts), the screw drives the nut to move horizontally, and the print 2 (nut connection) fixed by the nut drives the other four hard shafts to move horizontally in the same direction, so as to realize the telescopic function.

The main body of the material table is composed of a 20kg servo and 3D printed parts (including material placement area and driven cam). The servo is fixed on top of the lower print, and the upper and lower prints are connected by a chute. When the servo is energized, drive the cam to rotate to push the upper part of the print, and the print starts to slide under the push of the cam until it stops at the limit. The upper and lower prints are connected by large elastic rubber ropes or springs, which can keep the cam edge and the inside of the upper print always tangent, and realize the telescopic function of the material table.

C. Parameter setting

The geometric parameters of the handling robot are

mainly determined according to the volume of the robot, the size of the items to be grasped and the quality. When the volume parameters of the robot are controlled within $300 \times 300 \times 400 \text{ mm}^3$, it is necessary to reasonably design the size and stroke of the underbody plate and the robotic arm, expand the stroke as much as possible under the premise of meeting the requirements, expand the range that can be grasped, and carry out lightweight design to improve strength and reduce costs. In order to ensure the accuracy of the robotic arm and the robot as a whole and the stability of the transmission, the transmission device of the robotic arm must have a certain strength and stability, and the surface roughness of key parts must be controlled in a small range.

In order to facilitate the robot to move freely in a small range, it is necessary to design and select the motion mechanism and wheels under the car. After comprehensive consideration of various factors, Mecanum wheel is selected as the moving mechanism of the robot, on the one hand, with the stepper motor can achieve accurate motion positioning, on the other hand, because the structure and manufacturing process of the Mecanum wheel are more complex, it can bear a large load while ensuring omnidirectional movement. The specific parameters in Table I are designed within the volume parameter of $300 \times 300 \times 400 \text{ mm}^3$ and the accuracy error is within 3mm.

TABLE I GEOMETRIC PARAMETERS OF HANDLING ROBOTS

Parameter	Numeric value
Horizontal travel of the robotic arm L_f/mm	160
Vertical travel of the robotic arm L_h/mm	170
Robotic arm gimbal corner $\alpha/^\circ$	270
Material table travel $L_w(\text{mm} \times \text{mm})$	137
The gripper grips the maximum effective diameter D/mm	75
Chrome light rod 1 size $D_1 \times L_1/(\text{mm} \times \text{mm})$	6×227
Chrome light rod 2 size $D_2 \times L_2/(\text{mm} \times \text{mm})$	8×233
1204 profile length L_x/mm	233
Mecanum wheel size D_m/in	3
Upper and lower bottom plate dimensions $L_3 \times W_3/(\text{mm} \times \text{mm})$	266×251
Upper and lower bottom plate spacing W_1/mm	74.3
Overall dimensions in the folded state $L \times W \times H/(\text{mm} \times \text{mm} \times \text{mm})$	$300 \times 300 \times 400$
Total mass G/kg	8

D. Selection of Mecanum wheel and analysis of inverse kinematics model

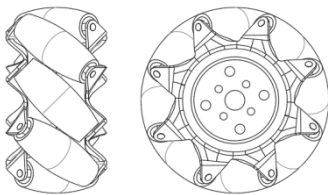


Fig. 2 Structural diagram of the Mecanum wheel



Fig. 3 Actual view of the Mecanum wheel

There are many sizes and models of Mecanum wheels on the market, and in this article, a 3-inch Mecanum wheel with nickel-plated steel and rubber rollers is selected, and is installed in an O-rectangular shape. The structure diagram is shown in Figure 2, and the physical diagram is shown in Figure 3. A single wheel has a total of 8 rollers, the angle between the axis of the roller and the axis of the wheel body is 45° , each roller is equipped with 2 bearings, and a single wheel has a total of 16 bearings. Each wheel weighs about 160g, and a group of (four) Mecanum wheels has a load bearing of about 40kg, which far exceeds that of other types of wheels of the same size.

Since each wheel is directly driven by a motor, the key to accurately controlling the movement of the robot requires calculating the steering and speed of each wheel, so a set of formulas is obtained by constructing an inverse kinematic model, which can solve the speed of each wheel according to the motion state of the chassis.

Objects in the plane have 3 degrees of freedom, that is, the motion in the two-dimensional plane can also be decomposed into three independent parameters, namely the translation of the X and Y axes and the rotation of the yaw

axis. This is shown in Figure 4.

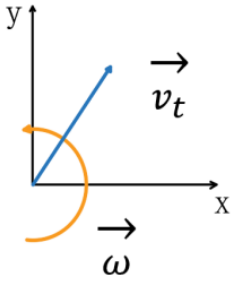


Fig. 4 Decomposition of planar motion

Decomposes the motion of the geometric center of each wheel into three quantities and defines:

The X-axis motion speed is $\mathcal{V}_{tx}$

The Y-axis motion speed is $\mathcal{V}_{ty}$

The angular velocity of Z-axis rotation is $\vec{\omega}$, Define counterclockwise as positive

The vector where the geometric center points to the axis of the wheel is $\vec{r}$

The velocity vector of the wheel axis is $\vec{\mathcal{V}}$

The wheel axis is perpendicular to $\vec{r}$ is $\vec{\mathcal{V}}_r$

Available:

$$\vec{\mathcal{V}} = \vec{\mathcal{V}}_t + \vec{\omega} \times \vec{r}$$

The components of the X and Y axes are calculated as

$$\begin{cases} \mathcal{V}_x = \mathcal{V}_{ty} - \mathcal{V}_{rx} = \mathcal{V}_{tx} - \omega \times r_y \\ \mathcal{V}_y = \mathcal{V}_{ty} + \mathcal{V}_{rx} = \mathcal{V}_{ty} + \omega \times r_x \end{cases}$$

This is shown in Figure 5

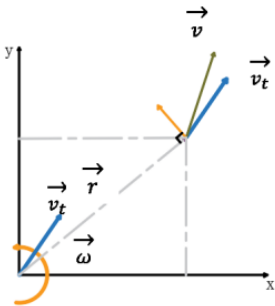


Fig. 5 Velocity decomposition of individual wheel shafts

The speeds of the other three wheel axes are shown in Figure 6:

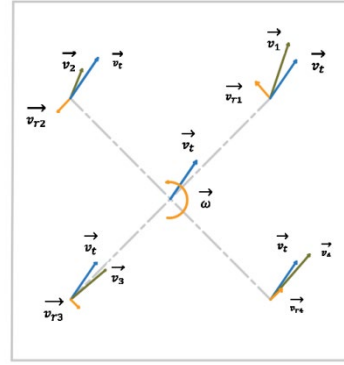


Fig. 6 Four-wheel motion breakdown

The velocity vector of a single wheel can be divided into two velocity components, which are a velocity vector perpendicular to the roller and a velocity vector parallel to the roller, because the roller axis is free to move, so the velocity vector perpendicular to the roller will directly act on the central axis of the roller, thereby providing the angular velocity of the roller rotating around the central axis, so this speed perpendicular to the roller has no effect on the trolley as a whole, as shown in Figure 7. Therefore, only attention needs to be paid to the velocity vector parallel to the rollers. And get the following formula

$$\mathcal{V}_{//} = \vec{\mathcal{V}} \cdot \vec{u} = (\mathcal{V}_x \vec{i} + \mathcal{V}_y \vec{j}) \cdot \left(-\frac{1}{\sqrt{2}} \vec{i} + \frac{1}{\sqrt{2}} \vec{j}\right) = -\frac{1}{\sqrt{2}} \mathcal{V}_x + \frac{1}{\sqrt{2}} \mathcal{V}_y$$

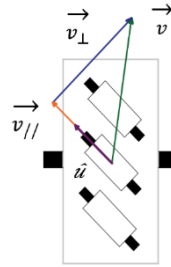


Fig.7 Roll motion decomposition

Calculation from roll motion to trolley RPM:

$$\mathcal{V}_{\omega} = \frac{\mathcal{V}_{//}}{\cos 45^{\circ}} = \sqrt{2} \left(-\frac{1}{\sqrt{2}} \mathcal{V}_x + \frac{1}{\sqrt{2}} \mathcal{V}_y\right) = -\mathcal{V}_x + \mathcal{V}_y$$

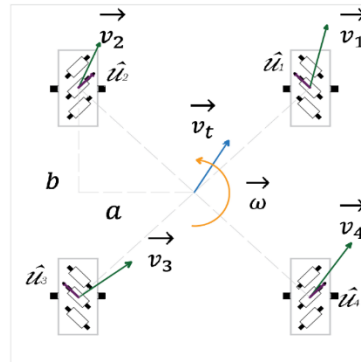


Fig. 8 Illustration of the overall motion

According to the definitions of a and b in Figure 8, we can obtain:

$$\begin{cases} \mathcal{V}_x = \mathcal{V}_{tx} + \omega b \\ \mathcal{V}_y = \mathcal{V}_{ty} - \omega a \end{cases}$$

Based on the above formula, the speed of the four wheels can be calculated according to the state of chassis movement:

$$\begin{cases} \mathcal{V}_{\omega 1} = \mathcal{V}_{ty} - \mathcal{V}_{tx} + \omega(a+b) \\ \mathcal{V}_{\omega 2} = \mathcal{V}_{ty} + \mathcal{V}_{tx} - \omega(a+b) \\ \mathcal{V}_{\omega 3} = \mathcal{V}_{ty} - \mathcal{V}_{tx} - \omega(a+b) \\ \mathcal{V}_{\omega 4} = \mathcal{V}_{ty} + \mathcal{V}_{tx} + \omega(a+b) \end{cases}$$

According to the above formula, the relationship between the motion state of the robot chassis and the steering and speed of each wheel can be solved.

III. MOTION ANALYSIS OF OMNIDIRECTIONAL MOBILE MATERIAL HANDLING ROBOTS

A. The three states and motion processes

The robot is divided into three forms: folded state, unfolded state and incomplete unfolded state, of which the incompletely unfolded state is the most flexible, and the volume of the robot can be selectively adjusted according to different scenarios to facilitate more efficient operation. The robot is in a folded state when it is idle, which can reduce the space occupied by the robot and improve the utilization rate of the site; When moving, it can be folded through a small space, making the robot move more flexible. The unfolded state can move a more selected stroke and grab items with a more selected distance, especially those on the inside of the high shelf.

1) Initial state robot (folded state)

When the robot is just started or not started, the robot is in the initial state (folded state), as shown in Figure 9, that is, at this time, the robot's mechanical claw shrinks at the minimum limit and points to the inside of the robotic arm, while the robotic arm as a whole is in the direction perpendicular to the four wheels, and the material table shrinks to the limit end in the direction of the gimbal, so that the robot as a whole is within the range of $300 \times 300 \times 400 \text{mm}^3$.



Fig. 9 Initial State Diagram

2) Unfolded state robot

When the robot is started or handled, if the space is wide (such as in a relatively empty room), the robot will change from a folded state to an unfolded state. As shown in Figure 10, the motor of the horizontal transmission mechanism of the robotic arm will act, and the synchronous wheel 1 fixed on the motor output shaft drives the synchronous wheel 2

fixed at the end of the screw to rotate through the synchronous belt, thereby driving the screw to rotate, and the nut and the light rod fixing on the screw will extend outward with the rotation of the screw, driving the middle four screws to extend horizontally. At the same time, the servo above the gripper begins to flip down 180° , and the rear gripper points in the opposite direction to the robotic arm. At the same time, the servo in the lower part of the material table also begins to operate, driving the cam to rotate so that the sliding seat of the material table extends outward, and finally stops at the limit of the outer end of the material table.

When the robot moves to the front of the material to be handled, the vision system identifies the center of the position where the material is located and forms two-dimensional coordinates, and then controls the robotic arm to extend until the robotic claw hovers accurately above the material (or 30mm in front), and then the robotic claw opens under the drive of the servo, while the robotic arm moves downward (or forward) until the material is within the grasping range of the robotic claw, and the servo on the robotic claw moves and grabs the material. After the grasping action is completed, the motor on the gimbal is connected to power and drives the manipulator to rotate to the material placement area behind the manipulator, and when the gimbal rotates, the manipulator extends out of the structure and retracts back, rotates to the material placement area and opens the claw to place the material.

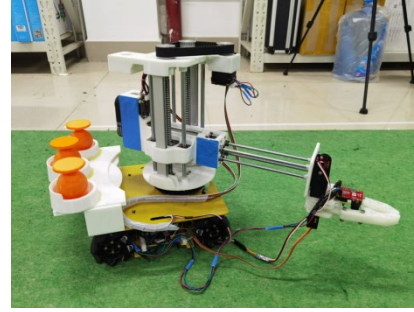


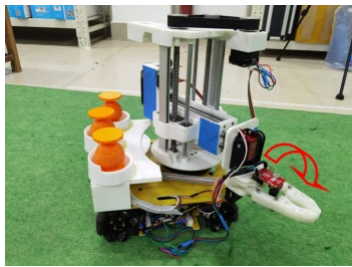
Fig. 10 Expanded State Diagram

The robot extension mechanism adopts a 4+4 combination, that is, 4 retractable hard axes + 4 guide hard shafts, a total of 8 chrome-plated hard shafts. Its diameter is 6mm, the length is 227mm, the material is bearing steel, it is electroplated and quenched, and the surface hardness can reach HRC602. When the mechanical claw extends out, it is equivalent to a cantilever structure, and the maximum bending moment occurs at the contact between the shaft and the linear bearing, that is, the linear bearing at the front end of the sliding seat, and the maximum theoretical load that the front end of the sliding seat can bear is 145kg, but affected by the processing method and the adhesion between the layers of materials of different manufacturers, the weight that can be grabbed by the front end of the robotic claw in the fully expanded state of the specimen robot has a large range of changes. When using stronger materials and better processing methods, a larger weight range can be grasped, but the gripping range is not unlimited, the gripping range is limited by both the torque of the mechanical jaw servo and the torque of the stepper motor, and the strength, stiffness and stability of the chrome-plated hard shaft far exceed other parts, so it can be ignored. Considering the cost and environmental protection issues, only the sliding seat made of 1.75mm ABS wire printed by FDM (3D) is used as an example for

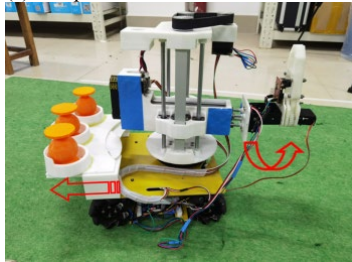
experimental analysis.

3) Partially expanded state robot

Incomplete unfolded state, as the name implies, is that some parts are still folded, or only partially expanded, such as when it is necessary to pass through a specific narrow channel, part of the joint needs to be folded, and the volume can be selectively reduced according to the specific environment while maintaining the flexibility of grasping and the range of grasping as much as possible. As shown in Figure 11 (a), the material table is folded, the paw is unfolded, and this state can enter the narrow channel to grab the item; While the material table is unfolded, the paw is half-folded and the order in which the material table is selectively placed, especially when the last few materials are in the middle and longer, as shown in Figure 11(b).



(a) The paw is unfolded and the material table is folded

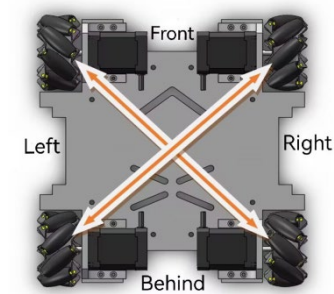


(b) The material table is unfolded and the paw is rotated

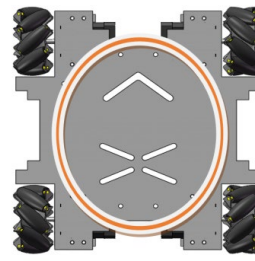
Fig. 11 Incomplete expansion state diagram

B. The motion scheme and movement process

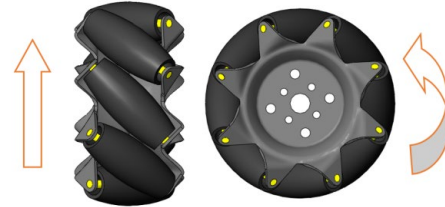
The McNum wheel is generally used in a group of four, including two left-handed and two right-handed wheels, and the installation methods are also diverse, but the commonly used is generally X-rectangular type and O-rectangular type installation (here refers to the direction of the upward view roller, that is, the part of the Maxnum wheel in contact with the ground is projected to the ground view), here the O-rectangular installation is used, and the direction of the underbody is directly forward, and the direction of wheel rotation is the direction of movement close to the side of the paper, as shown in Figure 12.



(a) O-mounted underbody top view

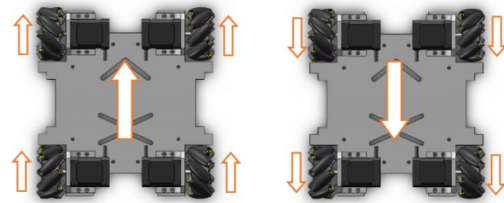


(b) O-type installation of the projection of the part of the car in contact with the ground

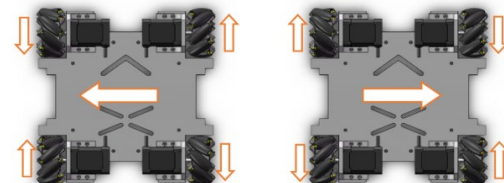


(c) Illustration of wheel steering and annotation direction

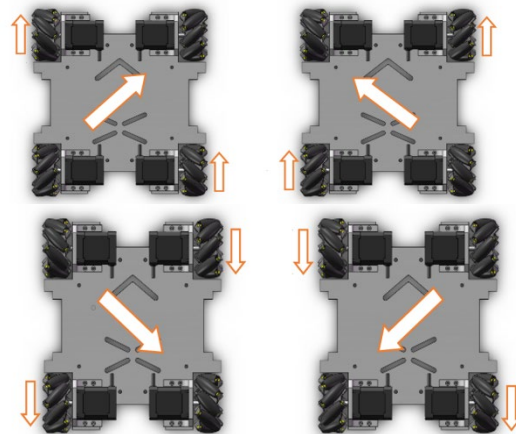
Fig.12 Schematic diagram of McClum wheel O-type installation



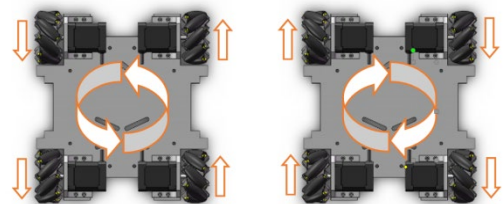
(a) Anteroposterior movement



(b) Left and right movement



(c) Oblique motion



(d) Turn in place

Fig.13 Illustration of wheel steering and robot movement direction

When the four wheels move forward at the same time, that is, the combined speed of each wheel $\vec{v}_1 = \vec{v}_2 = \vec{v}_3 = \vec{v}_4$. When it is the same and both are forward, the direction of movement of the robot is also forward, and forward movement can be achieved, and the speed of forward movement depends on $\vec{v}_i$'s size of the mold; Similarly, when the speed of the wheels closes $\vec{v}_1 = \vec{v}_2 = \vec{v}_3 = \vec{v}_4$ same and both backwards, The robot also moves backwards, as shown in Figure 13(a). While $\vec{v}_1 = -\vec{v}_2 = \vec{v}_3 = -\vec{v}_4$, left or right movement can be realized, that is, the left front wheel and the right rear wheel are backward movement, and the left rear wheel and right front wheel are forward movement, the left movement can be realized; Right-hand movement can be achieved when both the left front wheel and the right rear wheel are forward, and both the left rear wheel and the right front wheel are backward, as shown in Figure 13(b). When only two wheels are moving diagonally, and the direction of movement is the same, the oblique movement can be achieved, as shown in Figure 13(c). And when the two wheels on the left move backwards, and the two wheels on the right move forward, namely $\vec{v}_1 = -\vec{v}_2 = -\vec{v}_3 = \vec{v}_4$, It can realize the counterclockwise rotation of the robot in place; When both wheels on the left are forward and both wheels on the right are moving backward, namely $\vec{v}_1 = \vec{v}_2 = \vec{v}_3 = -\vec{v}_4$, it is possible to rotate the robot clockwise in place, as shown in Figure 13(d).

IV. PHYSICAL PRODUCTION AND FUNCTIONAL VERIFICATION OF OMNIDIRECTIONAL MOBILE MATERIAL HANDLING ROBOT

A. FDM fused deposition modeling 3D printing slice parameter setting

Additive manufacturing, also known as 3D printing, is a new and efficient rapid prototyping technology [13], through the import of three-dimensional digital model files, the use of filamentous linear, powder and liquid materials to create objects with complex shapes and diverse structures, which can save material loss in the manufacturing process and the cost of a variety of processing equipment [14-16]. FDM molding technology is currently the most widely used 3D printing technology because of its no pollution in the molding process, and low price of equipment and consumables, and its molding materials are numerous [17-18], of which acrylonitrile-butadiene-styrene (ABS) is a thermoplastic polymer consumable commonly used by FDM, which has the advantages of high impact resistance, corrosion resistance, easy molding and stable printing quality [19].

Because the design of some parts of the robot is more complex and unique, the processing cycle of other processing methods is long and the cost is high, so 3D printing technology is selected to make some parts. In order to obtain higher quality and better mechanical properties of the print, ABS with a wire diameter of 1.75mm is selected as a 3D printing consumable, and the printing layer thickness, top/bottom layer number, filling density and other aspects are studied and analyzed, and the printing trial is made. The Laservo A8 commercial-grade 3D printer is used for printing and production, and the Ultimaker Cura 5.0.0 software is used for slice settings. Taking the sliding seat of the robotic arm, which is also the largest part that the entire robot needs

to print, as an example, the printing layer thickness is 0.2mm, the top/bottom layer number is 4, the filling density is 60%, the filling pattern is straight, the printing temperature is 210°, and the platform temperature is 55°. A slice preview of layer 160 is shown in Figure 14.

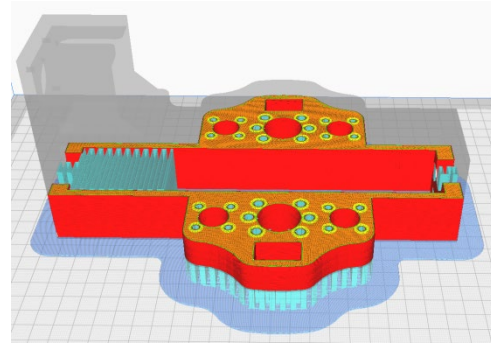


Fig.14 Slice parameter setting and checking

B. Strength check of sliding seats

In order to understand the force condition of the sliding seat during operation and the maximum load it can bear, it is necessary to carry out the relevant mechanical analysis of the sliding seat, which is analyzed here using finite element analysis software. The printing material used in the production of this slide seat is ABS, and the material parameters are shown in Table II below.

TABLE II MATERIAL PROPERTY TABLE

material	ABS			
attribute	Young's modulus	density	Poisson's ratio	Yield stress
	2000 MPa	1 060 kg/(m ³)	0.35	45 MPa

During the working process of the robotic arm, when the robot jaw extends to the maximum stroke, the load of the sliding seat will reach the maximum. Considering that the mechanism of the robotic arm is relatively complex, and the sliding seat bears the role of multiple loads at the same time, it is inconvenient to simplify the load, so the assembly is used for analysis. According to the actual working conditions, the fixed constraint is applied to the connection between the sliding seat and the screw, because the guide light rod does not have a limiting effect on the vertical movement of the sliding seat, so the displacement constraint is applied at the sliding seat and the guide light rod, and the degree of freedom in the vertical direction is in a free state. The specific structure and the distribution of boundary conditions are shown in Figure 15.

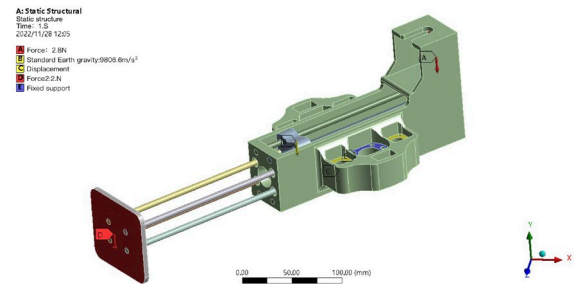


Fig.15 Loads and boundary conditions

In the process of solving, the constraints and loads are usually kept unchanged, the mesh is gradually refined, the model is calculated, the calculation results under different

mesh conditions are compared, and the irrelevance of the results and the mesh is judged. The analysis is carried out by encrypting the grid step by step, and the analysis results are shown in Table III and the stress changes are shown in Figure 16.

TABLE III RESULTS OF GRID AGNOSTICITY ANALYSIS

Unit size /mm	Number of units	stress /mpa	Relative increment of stress
6	26250	0.229	
5	38333	0.215	6.51%
4	64589	0.232	7.33%
3	135119	0.234	0.855%

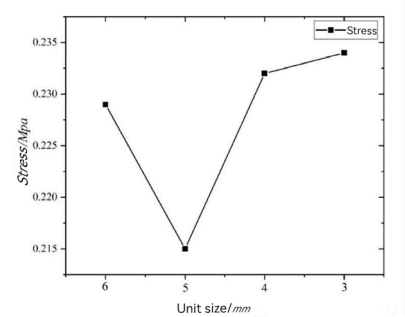
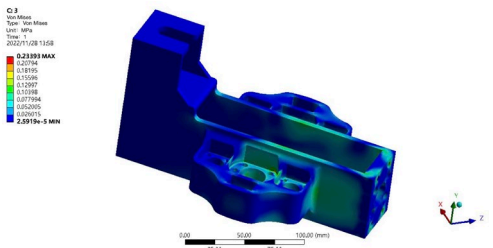
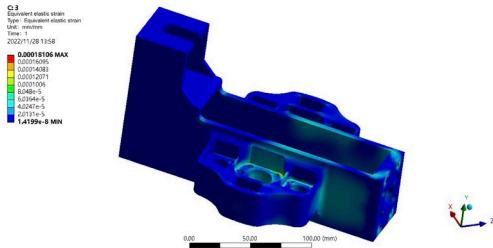


Fig.16 Stress change with element size

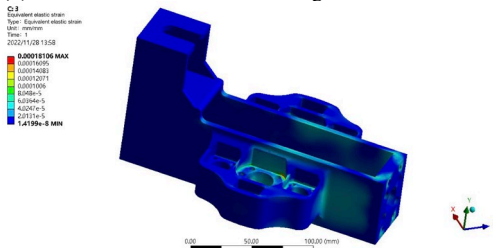
According to the above results, when the cell size is refined from 4mm to 3mm, the change in stress value is the smallest, controlled at 0.855%. Therefore, when performing finite element static analysis, the grid element size is 3 mm. The stress analysis results of the slide seat are shown in Figure 17.



(a) Slide seat isometric force cloud



(b) Slide seat elastic strain cloud diagram



(c) Slide seat deformation cloud

Fig.17 Results of static analysis

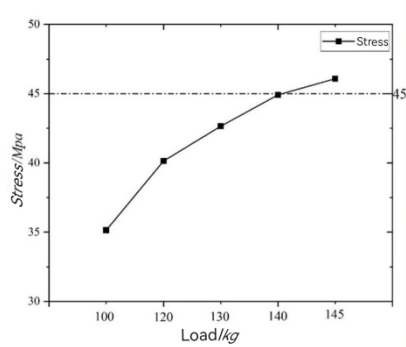


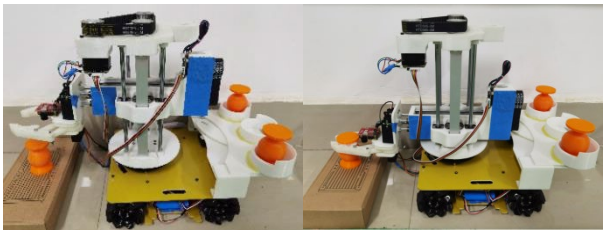
Fig.18 Stress change with load

It can be seen from Figure 18 that as the load of the manipulator part gradually increases, the stress borne by the manipulator arm and the strain of the manipulator also gradually increase, when the load gradually increases to 145kg, the manipulator slide seat is damaged, reaching the maximum stress that the ABS material can withstand the sliding seat, at this time the maximum stress of the manipulator is 46.08MPa. However, considering that the parts printed by FDM technology are affected by their molding methods, the maximum value of the actual stress that can withstand is less than 46.08MPa, but it is still much greater than the load generated by the actual transported items, so the structural design scheme of the robot meets the strength requirements.

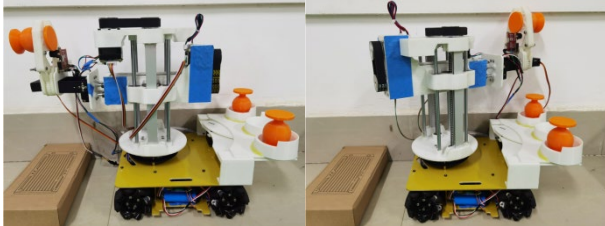
C. Physical production and functional verification

The high flexibility of omnidirectional mobile material handling robots that can adapt to complex environments allows them to grasp and transport materials in complex environments.

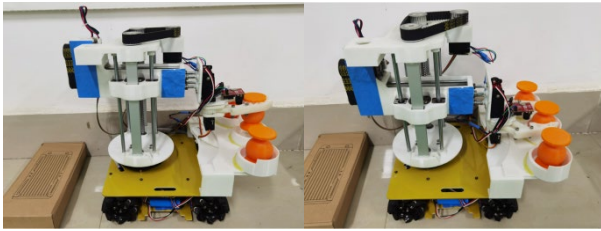
Figure 19 shows the process of action and grabbing by the robotic arm when the material is handled high and the order in which the materials are placed on the material table is required: (1) OpenMV recognition (2) Robotic claw grasping (3) The robotic arm rises while the robotic claw rotates (4) The gimbal rotates (5) The manipulator rotates in the opposite direction (6) The manipulator descends and releases the claw (7) The manipulator rises, and the robot moves the material.



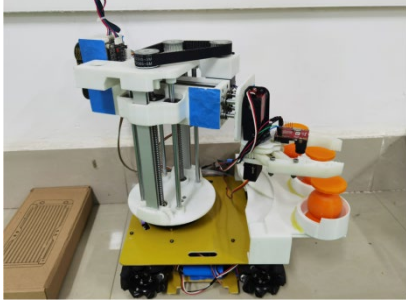
(a) OpenMV Identification (b) Gripping by mechanical claws



(c) The robotic arm rises while the robotic claw rotates(d) Gimbal rotation



(e) The gripper rotates in reverse (f) The robotic arm descends and releases the claws



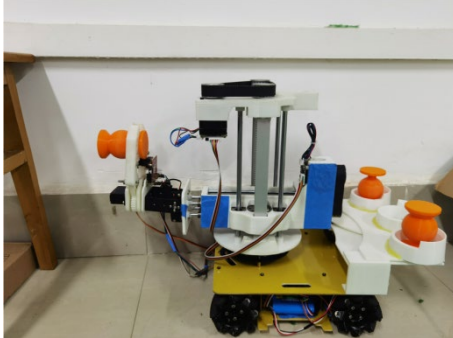
(g) The robotic arm rises and the robot moves the material

Fig. 19 Flow diagram of the handling robot grabbing the material

Figure 20 shows the action process of the robotic arm to grasp the narrow compartment material: (1) the robotic arm extends (2) identifies and grasps (3) the robotic claw is reversed upwards and the materials are placed sequentially



(a) Robotic arm extended (b) Identify and Scrape



(c) The gripper is reversed upwards and placed in order

Fig. 20 Narrow compartment material grab handling

Figure 21 shows the process of the robot identifying and grasping the material in the narrow lane (only the key steps are given here, other steps are similar to Figure 18) : (1) the robot recognizes and grasps (2) the robot claw recycling



(a) Robot identification and grasping (b) Robotic claw recycling

Fig. 21 The robot in the narrow lane grabs and handles the material

V. CONCLUSION

This paper introduces in detail the structural design and motion process of omnidirectional mobile material handling robot that can adapt to complex environment, analyzes the inverse kinematics model and application prospect of Mecanum wheel in movable handling robot, improves the problem of obstruction of the movement of conventional robotic arm in complex handling environment, improves the flexibility of handling robot, and reduces the overall volume of the robot and improves the utilization rate of idle space.

Finally, through the grasping and transportation experiments of the handling robot, it is shown that the robot can grasp, handle and place experiments through various complex environments, indicating that the robot has good practicality. Through the theoretical model and stress analysis of the handling robot, the accurate control and smooth operation of the robot can be realized, so that it can more efficiently and better complete the task of grasping and transportation in complex environments.

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